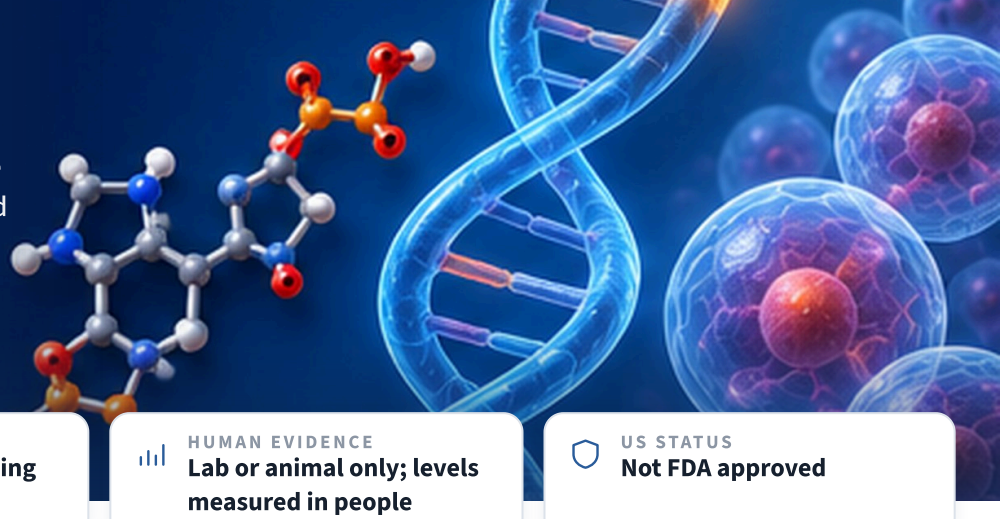
LONGEVITY 12

Klotho

Protein made by the body; one form sits in the cell membrane and another circulates in blood



CLASS	STUDIED FOR	HUMAN EVIDENCE	US STATUS
Protein	Lifespan, memory, aging	Lab or animal only; levels measured in people	Not FDA approved

STUDIED FOR

Memory	Lifespan	Newborn lungs
ANIMAL	ANIMAL	ANIMAL

Protein never given to people. Human data link blood levels or gene variants with health. Two small klotho gene-therapy pilots have posted no results.

HOW IT WORKS IN THE BODY

Klotho → Circulating form → Insulin/IGF-1 → Aging in mice

- **Mice lacking klotho** aged early and died young (1997)
- **Circulating klotho** dampened insulin and IGF-1 signalling, in a mouse study
- **One injection** improved memory tests in old monkeys at a low dose only
- **In people,** the mechanism is not established

EVIDENCE & WHAT'S KNOWN

EVIDENCE

LAB OR ANIMAL
LEVEL 1 OF 5

- Mouse, rat and one monkey study
- No study gave klotho protein to people
- 2 gene-therapy pilots, no results

NOT YET ESTABLISHED

- Safety in people
- Half-life in people
- Dose in people
- Effects of higher doses

QUICK PRACTICAL INFO

HALF-LIFE	ROUTES STUDIED
Not established in people	None studied in people (protein)
PHARMACOKINETICS	**SOLD AS**
Not established in people	"Research use only" (not for human use)
MONITORED IN TRIALS	**SPORT (WADA)**
Not established (protein never given to people)	Not named on the 2026 list

REGULATORY	FDA	WADA
	Not approved; not on compounding lists	Not named; S0 bans unapproved substances

REFERENCES

1. Nature Aging, klotho injection and memory in old rhesus monkeys, Jul 2023
2. Science, suppression of aging in mice by the hormone klotho, Sep 2005
3. WADA, 2026 Prohibited List

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